CEFET/RJ

DAL & Comunidade

# PACOTES E AJUDA

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# *Pacotes*

- Pacotes são coleções de funções disponibilizados como bibliotecas
  - Publicados no repositório CRAN
    - Controle de qualidade
    - Boa documentação
  - Carregados de um repositório GitHub
    - Versões em desenvolvimento
- Atualmente têm mais de 18500 pacotes
  - https://cran.r-project.org/
- R tem uma comunidade muito ativa
  - Vários pesquisadores, professores, programadores e estatísticos
- Três pacotes são do Data Analytics Lab
  - https://cran.r-project.org/web/packages/TSPred/index.html
  - https://cran.rstudio.com/web/packages/gstsm/index.html
  - https://cran.r-project.org/web/packages/STMotif/index.html

# *Instalação de pacotes*

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```
#funções gráficas qplot
?qplot

library(ggplot2)

?qplot

install.packages("ggplot2")
```

8:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from C:/R/main/.RData]

> ?qplot
No documentation for 'qplot' in specified packages and libraries:
you could try '??qplot'
> library(ggplot2)
> ?qplot
>
```

Environment History Connections Tutorial

Import | 132 MiB | List

R | Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

R: Quick plot | Find in Topic

qplot {ggplot2} R Documentation

## Quick plot

### Description

Deixe comentado o install.packages para não ficar rodando várias vezes a instalação

# *Uso de função do pacote ggplot2*

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R

Source on Save | Run | Source

```r
x <- c(-1, -0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8, 1)
x

y <- x^3
y

qplot(x, y)
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> x <- c(-1, -0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8, 1)
> x
 [1] -1.0 -0.8 -0.6 -0.4 -0.2  0.0  0.2  0.4  0.6  0.8  1.0
>
> y <- x^3
> y
 [1] -1.000 -0.512 -0.216 -0.064 -0.008  0.000  0.008  0.064
 [9]  0.216  0.512  1.000
>
> qplot(x, y)
>
```

Environment History Connections Tutorial

Import | 178 MiB | List

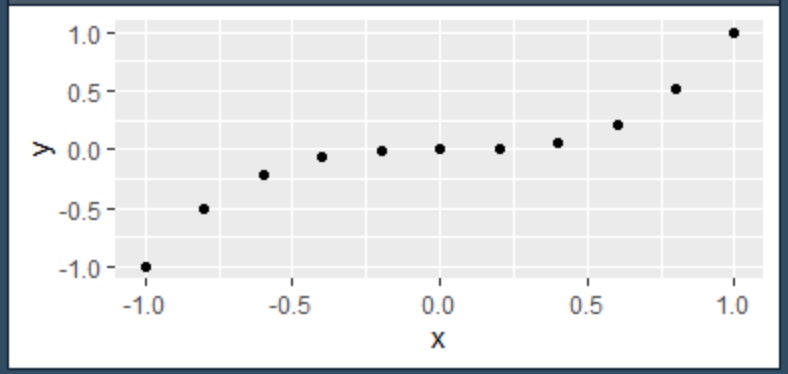
R | Global Environment

values

| | |
|---|---|
| x | num [1:11] -1 -0.8 -0.6 -0.4 ... |
| y | num [1:11] -1 -0.512 -0.216 -... |

Files Plots Packages Help Viewer Presentation

Zoom | Export




Chamadas com parâmetros explícitos não precisa seguir a ordem os argumentos

# *Prototipando uma função*

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R

Source on Save | Run | Source

```
1
2  # criar uma função para jogada
3  dado <- 1:6
4  dados <- sample(dado, size = 2, replace = TRUE)
5  sum(dados)
6
7
8
9
```

8:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> source("C:/Users/eduar/OneDrive/Aulas/R/02-Pacotes-Ajuda.R")
>
```

Environment History Connections Tutorial

Import 191 MiB List

R Global Environment

values

| | |
|---|---|
| dado | int [1:6] 1 2 3 4 5 6 |
| dados | int [1:2] 4 5 |

Files Plots Packages Help Viewer Presentation

C: > R > main

| Name | Size | Modified |
|---|---|---|
| .. | | |
| .RData | 122 B | Sep 19, |
| .Rhistory | 26.5 KB | Sep 27, |

# *Prototipando uma função*

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```
1  jogada <-  function() {
2    dado <- 1:6
3    dados <- sample(dado, size = 2, replace = TRUE)
4    return(sum(dados))
5  }
6
7  jogada()
```

1:1 | jogada() | R Script

Console | Terminal | Background Jobs

R 4.2.1 · C:/R/main/

```
> source("C:/Users/eduar/OneDrive/Aulas/R/02-Pacotes-Ajuda.R")
> jogada <-  function() {
+   dado <- 1:6
+   dados <- sample(dado, size = 2, replace = TRUE)
+   return(sum(dados))
+ }
>
> jogada()
[1] 7
>
```

Environment | History | Connections | Tutorial

Import | 191 MiB | List

R | Global Environment

Functions

| | |
|---|---|
| jogada | function () |

Files | Plots | Packages | Help | Viewer | Presentation

C: > R > main

| Name | Size | Modified |
|---|---|---|
| .. | | |
| .RData | 122 B | Sep 19, |
| .Rhistory | 26.5 KB | Sep 27, |

Criando a função

Clique na vassoura para limpar as variáveis

# *Simulação "mínima"*

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```
simulacao_minima <- replicate(30, jogada())
qplot(simulacao_minima, binwidth =1)
mean(simulacao_minima)
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> simulacao_minima <- replicate(30, jogada())
> qplot(simulacao_minima, binwidth =1)
> mean(simulacao_minima)
[1] 6.533333
>
```

Environment History Connections Tutorial

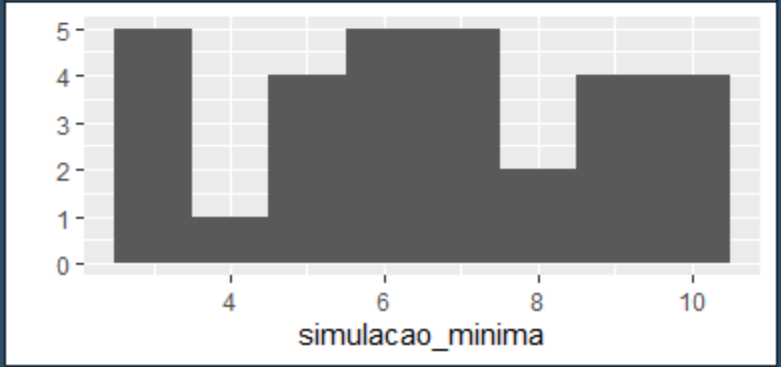
Import | 187 MiB | List

R | Global Environment

| Values | |
|---|---|
| simulacao_... | int [1:30] 10 7 6 10 6 8 3 6 ... |
| **Functions** | |
| jogada | function () |

Files Plots Packages Help Viewer Presentation

Zoom | Export




O número mínimo de amostras para calcular uma média é 30, mas, pode não ser suficiente para apresentar uma distribuição ...

# *Espaço de possibilidades*

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | (6,1) | | | | | |
| combinations | | | | | (5,1) | (5,2) | (6,2) | | | | |
| of die 1 and | | | | (4,1) | (4,2) | (4,3) | (5,3) | (6,3) | | | |
| die 2 that add | | | (3,1) | (3,2) | (3,3) | (3,4) | (4,4) | (5,4) | (6,4) | | |
| up to sum | | (2,1) | (2,2) | (2,3) | (2,4) | (2,5) | (3,5) | (4,5) | (5,5) | (6,5) | |
| | (1,1) | (1,2) | (1,3) | (1,4) | (1,5) | (1,6) | (2,6) | (3,6) | (4,6) | (5,6) | (6,6) |
| | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** |
| | | | | | | **sum** | | | | | |

Veja o cenário das possibilidades (probabilidades)
https://rstudio-education.github.io/hopr/packages.html

# *Simulação "completa"*

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```
simulacao_completa <- replicate(10000, jogada())
qplot(simulacao_completa, binwidth =1)
mean(simulacao_completa)
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> simulacao_completa <- replicate(10000, jogada())
> qplot(simulacao_completa, binwidth =1)
> mean(simulacao_completa)
[1] 7.0006
>
```

Environment History Connections Tutorial

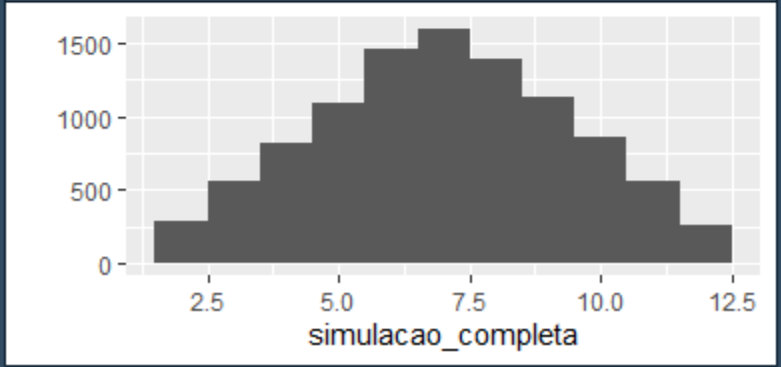
Import | 186 MiB | List

R | Global Environment

| Values | |
|---|---|
| simulacao_... | int [1:10000] 6 5 4 7 8 10 4 ... |
| simulacao_... | int [1:30] 10 7 6 10 6 8 3 6 ... |
| **Functions** | |
| jogada | function () |

Files Plots Packages Help Viewer Presentation

Zoom | Export



# *Ajuda para raiz quadrada*

R: Miscellaneous Mathematical Functions  Find in Topic

MathFun {base}  R Documentation

## Miscellaneous Mathematical Functions

### Description

`abs(x)` computes the absolute value of `x`, `sqrt(x)` computes the (principal) square root of `x`, $\sqrt{x}$.

The naming follows the standard for computer languages such as C or Fortran.

### Usage

```
abs(x)
sqrt(x)
```

### Arguments

`x` a numeric or `complex` vector or array.

### Details

These are internal generic primitive functions: methods can be defined for them individually or via the `Math` group generic. For complex arguments (and the default method), `z, abs(z) == Mod(z)` and `sqrt(z) == z^0.5`.

`abs(x)` returns an `integer` vector when `x` is `integer` or `logical`.

### S4 methods

Both are S4 generic and members of the `Math` group generic.

### References

Becker, R. A., Chambers, J. M. and Wilks, A. R. (1988) *The New S Language*. Wadsworth & Brooks/Cole.

### See Also

`Arithmetic` for simple, `log` for logarithmic, `sin` for trigonometric, and `Special` for special mathematical functions.

'plotmath' for the use of `sqrt` in plot annotation.

### Examples

Run examples

```
require(stats) # for spline
require(graphics)
xx <- -9:9
plot(xx, sqrt(abs(xx)),  col = "red")
lines(spline(xx, sqrt(abs(xx)), n=101), col =
```

[Package *base* version 4.2.1 Index]

?sqrt

# *Usando o código da ajuda*

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R

Source on Save | Run | Source

```
1 require(stats) # for spline
2 require(graphics)
3 xx <- -9:9
4 plot(xx, sqrt(abs(xx)),  col = "red")
5 lines(spline(xx, sqrt(abs(xx)), n=101), col =  "pink")
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> require(stats) # for spline
> require(graphics)
> xx <- -9:9
> plot(xx, sqrt(abs(xx)),  col = "red")
> lines(spline(xx, sqrt(abs(xx)), n=101), col =  "pink")
>
```

Environment History Connections Tutorial

Import | 194 MiB | List

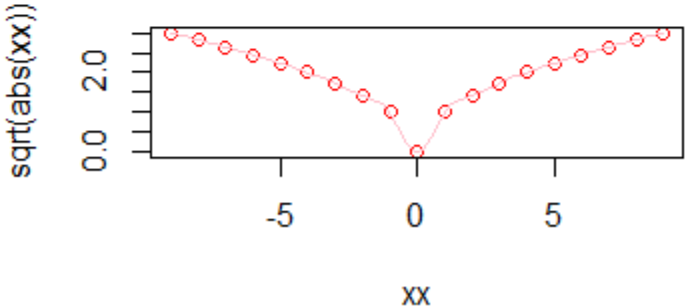
R | Global Environment

Values

| | |
|---|---|
| xx | int [1:19] -9 -8 -7 -6 -5 -4 ... |

Files Plots Packages Help Viewer Presentation

Zoom | Export




Explore os exemplos da documentação

# *Ajuda para amostra*

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```
1 ?sample
2
```

2:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> require(stats) # for spline
> require(graphics)
> xx <- -9:9
> plot(xx, sqrt(abs(xx)),  col = "red")
> lines(spline(xx, sqrt(abs(xx)), n=101), col =  "pink")
> ?sample
>
```

Environment History Connections Tutorial

Import | 195 MiB | List

R | Global Environment

Values

| | |
|---|---|
| xx | int [1:19] -9 -8 -7 -6 -5 -4 … |

Files Plots Packages Help Viewer Presentation

R: Random Samples and Permutations | Find in Topic

| | |
|---|---|
| x | either a vector of one or more elements from which to choose, or a positive integer. See 'Details.' |
| n | a positive number, the number of items to choose from. See 'Details.' |
| size | a non-negative integer giving the number of items to choose. |
| replace | should sampling be with replacement? |
| prob | a vector of probability weights for obtaining the elements of the vector being sampled. |
| useHash | logical indicating if the hash-version of the |

# *Versão da amostragem com distribuição de probabilidades*

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

02-Pacotes-Ajuda.R*

Source on Save | Run | Source

```r
simulacao = replicate(10000,
                      sample(2:12,
                      replace=TRUE,
                      prob=c(1,2,3,4,5,6,5,4,3,2,1)/36))
qplot(simulacao, binwidth=1)
mean(simulacao)
```

6:16 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> simulacao = replicate(10000,
+                       sample(2:12,
+                       replace=TRUE,
+                       prob=c(1,2,3,4,5,6,5,4,3,2,1)/36))
> qplot(simulacao, binwidth=1)
> mean(simulacao)
[1] 7.000945
>
```

Environment History Connections Tutorial

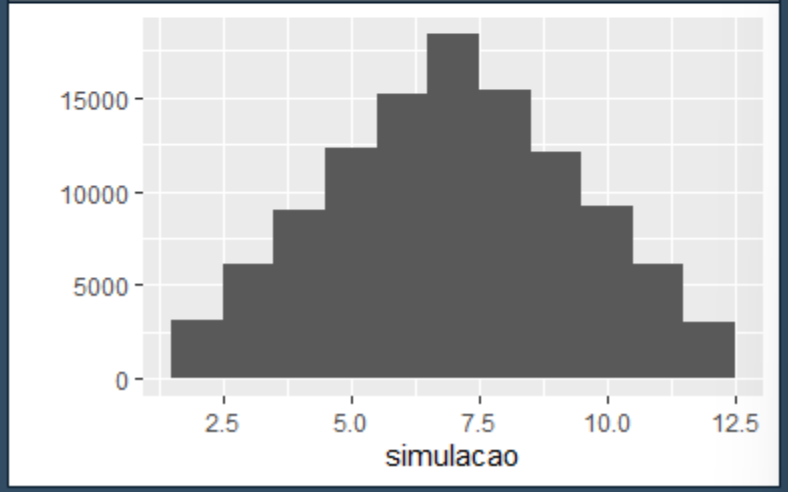
Import | 334 MiB | List

R | Global Environment

| Data | |
|---|---|
| simulacao | int [1:11, 1:10000] 7 8 9 5... |
| **Values** | |
| xx | int [1:19] -9 -8 -7 -6 -5 -4 ... |

Files Plots Packages Help Viewer Presentation

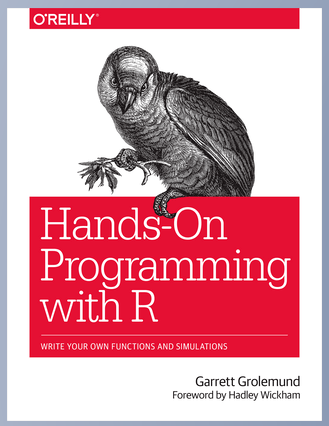
Zoom | Export

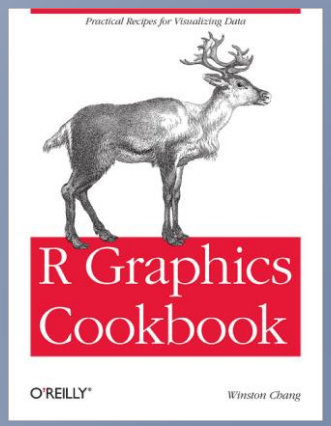


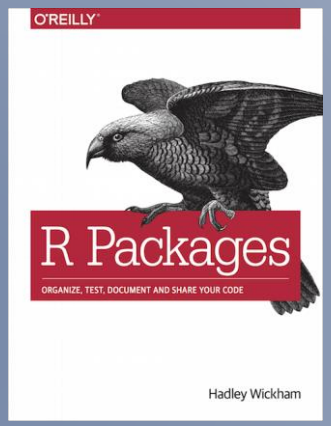

Usando a distribuição de probabilidade para cada valor possível

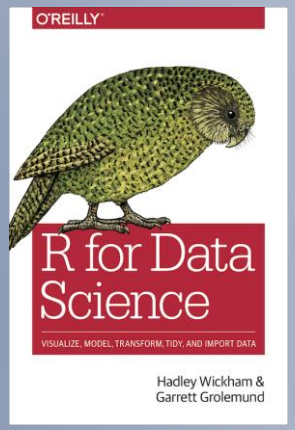
# *Referências*

Código: https://nbviewer.org/github/eogasawara/R/blob/main/02-Pacotes-Ajuda.ipynb



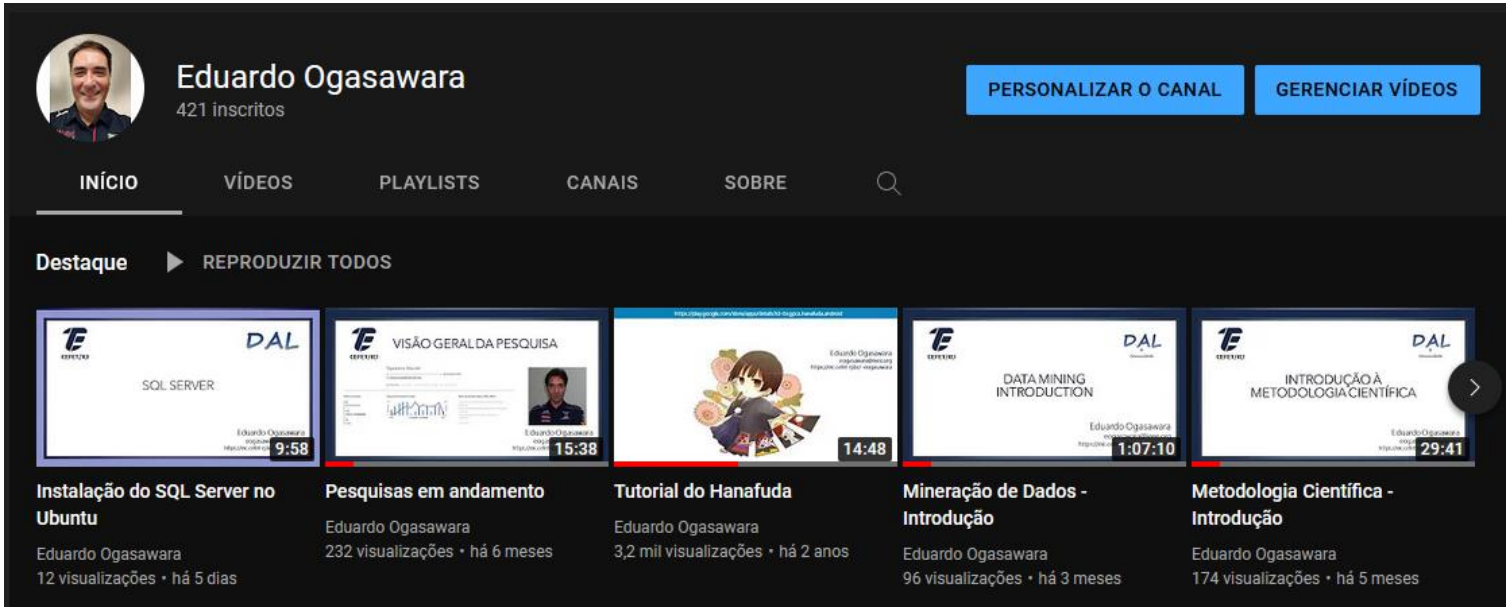

Hands-on Programming with R: https://rstudio-education.github.io/hopr/index.html
R Graphics Cookbook: https://r-graphics.org
R Packages: https://r-pkgs.org/index.html
R for Data Science: https://r4ds.had.co.nz

Leitura adicional: https://rstudio-education.github.io/hopr/packages.html

[1] G. Grolemund, 2014, Hands-On Programming with R: Write Your Own Functions and Simulations. O'Reilly Media, Inc.

# Mantenha-se atualizado

Inscreva-se: https://eic.cefet-rj.br/~eogasawara/youtube




Slides and Jupyter notebooks: https://github.com/eogasawara/R